

SOC Alert Simulation Report

Objective :

To simulate and investigate SSH authentication activity using Kali Linux logs and identify potential security events.

Tools Used :

- Kali Linux
- OpenSSH Server
- journalctl
- grep
- Terminal

Commands Used :

ifconfig

ssh kali@10.0.2.15

journalctl | grep ssh

journalctl | grep password

journalctl | grep login

Steps Performed :

1. Installed and started the SSH service in Kali Linux.
2. Identified the local IP address using ifconfig.
3. Connected to the system remotely using SSH.
4. Accepted the SSH host authenticity prompt.
5. Logged into the Kali system using SSH credentials.
6. Investigated generated logs using journalctl and grep commands.

Observations :

- SSH authentication activity was successfully generated.
- Remote login session was established from IP address 10.0.2.15.
- System logs recorded authentication and login events.

Findings :

- SSH service was active and accessible.
- Authentication logs were successfully generated and monitored.
- Login events appeared in system logs and can be used for SOC investigations.

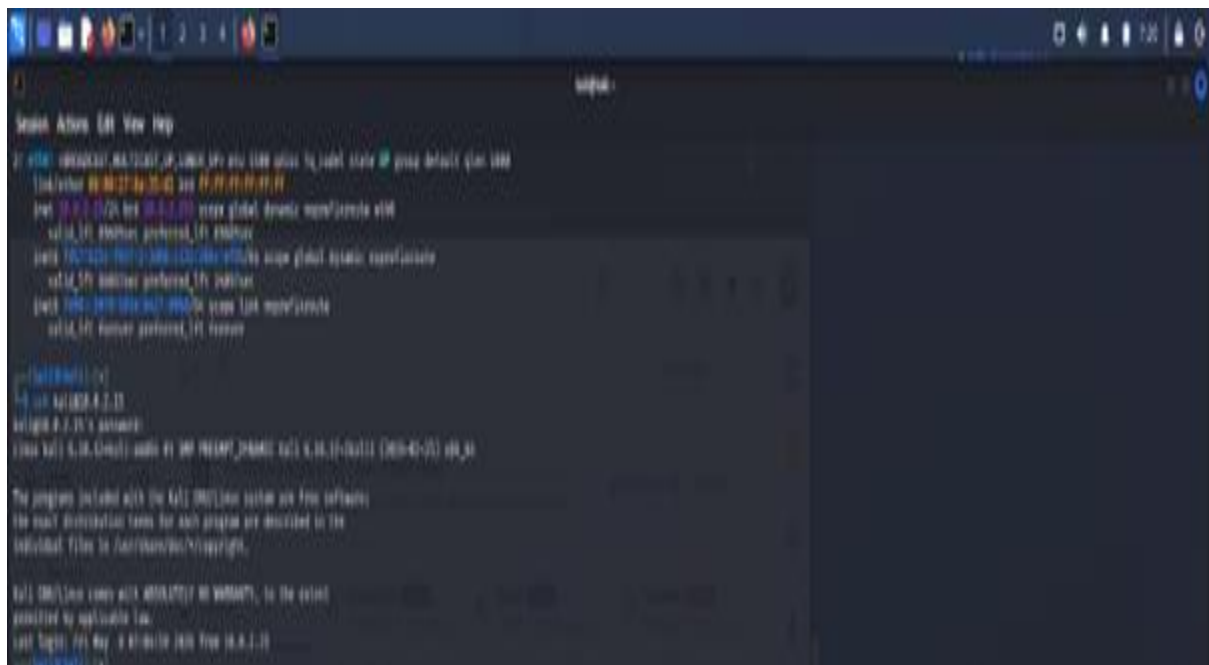
Conclusion :

The SOC alert simulation successfully demonstrated how authentication activities generate logs that can be monitored and analyzed by SOC analysts. SSH login monitoring is an important part of threat detection and incident investigation.

Recommendations

- Monitor SSH authentication logs continuously.
- Restrict unauthorized SSH access.
- Use strong passwords and multi-factor authentication.
- Configure SIEM tools for automated alert generation.

Screenshots :



```
ubuntu@ubuntu:~$ cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
ubuntu:x:1000:1000:ubuntu:/home/ubuntu:/bin/bash
ubuntu-bionic:x:1001:1001:ubuntu-bionic:/home/ubuntu-bionic:/bin/bash
```

[illegible]